Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

Fall Semester 2026-27

BCSE331P – Exploratory Data Analysis

Assessment 2

Experiment 4: Data Import, Transform and Basic Statistical Analysis CO2 BT2

CO1: Engrave simple R programs.

Write an R program to import the given dataset and perform preliminary data exploration.

1. Import the given dataset (CSV/Excel) into R. (Dataset share through email)
2. Display the first 10 records of the dataset.
3. Display the last 10 records of the dataset.
4. Find the total number of rows and columns.
5. Display the names of all variables (columns).
6. Display the data type (structure) of each variable.
7. Display the summary statistics of the dataset.
8. Determine the number of missing values in each column.
9. Display the total number of missing values in the dataset.
10. Display the number of missing values in each column.
11. Identify the rows containing at least one missing value.
12. Remove all rows containing missing values and display the cleaned dataset.
13. Replace missing values in a selected numerical column using the mean of that column.
14. Replace missing values in another numerical column using the median.
15. Replace missing values in a categorical column using the mode.
16. Perform forward filling to replace missing values with the previous non-missing value. (Load the package zoo, and use the function locf())
17. Perform backward filling to replace missing values with the next non-missing value.
18. Replace remaining missing values with a constant value (0 or "Unknown").
19. Remove duplicate records from the dataset.
20. Convert appropriate variables into factors.
21. Rename one or more column names.
22. Create a new derived column using existing variables.
23. Display the summary statistics of the dataset using summary().
24. Find the minimum and maximum values.
25. Calculate the range of a numerical variable.
26. Calculate the variance.
27. Calculate the standard deviation.
28. Calculate the interquartile range (IQR).

Note:

Don’t display all the record, if more than 10 records from your dataframe are displayed, then you can use head command